

A CRM APPLICATION FOR LAPTOP RENTALS

1. Project Overview:

This project is dedicated to developing a robust CRM application tailored for managing laptop rentals within a business environment using the Salesforce platform. The primary goal is to tackle operational challenges, including handling rental requests efficiently, monitoring inventory, and facilitating effective communication with customers. Leveraging Salesforce's strengths in automation, workflow optimization, and data storage, this CRM application aims to:

- a. Optimize the Rental Workflow: Streamline the booking and return processes to reduce time and minimize manual intervention required for each rental.
- b. Enhance Customer Satisfaction: Send timely email notifications to customers for booking confirmations and return reminders.
- c. Increase Data Accuracy and Insightful Reporting: Provide real-time visibility into laptop inventory and rental status, empowering staff to make informed, quick decisions.

2. Objective Goals:

- d. Streamline the laptop rental and return processes to minimize manual effort, enhancing efficiency and accuracy.
- e. Strengthen customer management practices to improve service quality and foster lasting customer relationships.
- f. Implement precise tracking and reporting for laptop inventory to ensure accurate availability and effective asset management.

Specific outcomes:

- i. A customized Salesforce CRM application designed to manage rental services with an intuitive interface, allowing users to easily view, manage, and track rental activity.
- ii. Automated workflows streamline the handling of rental requests, provide real-time status updates, and send email notifications.

- iii. Comprehensive reports and dashboards deliver real-time insights into inventory levels, rental activity, and customer engagement.

3. Salesforce Key Features and Concepts Utilized:

This section details the core Salesforce functionalities and tools used in the Laptop Rentals CRM application.

- **Custom objects and fields:**
 - a. **Laptop_Bookings__c:** A custom object created to manage each rental booking, featuring fields such as:
 - i. Customer Name
 - ii. Email__c
 - iii. Amount__c
 - iv. Core__c
 - v. Laptop_Type__c
 - vi. Status
 - vii. Laptops__c: An additional custom object representing inventory items, which tracks details like model, specifications, and availability status

Apex Triggers and Classes:

- a. Developed an after-insert trigger (LaptopBooking) to send automated confirmation emails when a booking is made. The LaptopBookingHandler class formats and personalizes these emails with customer details.

Process Automation with Process Builder and Flows:

- b. Designed a Process Builder flow to manage booking status changes (e.g., from "Booked" to "In Use" to "Returned") without manual updates.

Validation Rules for Data Accuracy:

- c. Added rules to enforce data quality, requiring key fields like Email and Amount for complete booking records.

Reports and Dashboards: Created dashboards and reports to provide insights on **inventory, rental volumes, popular laptop models**, and customer demographics, supporting strategic

planning and resource allocation.

4. Detailed Steps to Solution design:

The solution was carefully designed in multiple steps, covering data models, user - interface, and business logic. Here's an overview:

- Data Model Design:
 - a. Created **Laptop_Bookings__c** and **Laptops__c** custom objects.
 - b. Established relationships between **Laptop_Bookings__c** and standard Salesforce objects (Account or Contact) to connect customers with their respective bookings.
 - c. Additional fields, such as  Email__c, Amount__c, Core__c, and Laptop_Type__c, are used to store booking-specific data. Relationships between objects are established to link the Laptop_Bookings__c object with other standard or custom objects (e.g., Account, Contact) to enhance data integrity and support comprehensive reporting.

Fields Included in the **LaptopBooking** object:

Setup	Home	Object Manager
SETUP > OBJECT MANAGER		
Laptop Bookings		
Details	Fields & Relationships	
Fields & Relationships	11 Items, Sorted by Field Label	
Page Layouts	Quick Find New Deleted Fields Field Dependencies Set History Tracking	
Lightning Record Pages		
Buttons, Links, and Actions		
Compact Layouts		
Field Sets		
Object Limits		
Record Types		
Related Lookup Filters		
Search Layouts		
List View Button Layout		
Restriction Rules		
Scoping Rules		
	FIELD LABEL	FIELD NAME
	DATA TYPE	CONTROLLING FIELD
	INDEXED	
	Amount	Amount__c
	Currency(18, 0)	
	Consumer	Consumer__c
	Master-Detail(consumer)	
	Core Type	Core_Type__c
	Picklist	Laptop Names
	Created By	CreatedById
	Lookup(User)	
	Email	Email__c
	Email	
	how many months	how_many_months__c
	Picklist	
	Laptop Bookings	Name
	Text(80)	
	Laptop Names	Laptop_Names__c
	Picklist	
	Laptops Available	Laptops_Available__c
	Formula (Number)	

User Interface (UI) Design:

The custom Lightning app **Laptop Rentals** features easy navigation with tabs for key components, including **Bookings**, **Reports**, and **Dashboards**.

1. Configured Page Layouts for **Laptop_Bookings__c** and **Laptops__c** with relevant fields, sections, and related lists for ease of access.
2. Designed dashboards for quick visual insights on active rentals, availability and top-performing laptops.
3. Custom Lightning components may be added to enhance specific sections, such as displaying a chart of most rented laptop models.

Navigation Items on Laptop Rental Application:

LAPTOP RENTALS Total Laptops consumer Laptop Bookings Billing Process Laptop Analytics data analytics of laptops

Search...

New Import Change Owner Assign Label

consumer Recently Viewed

9 items • Updated a few seconds ago

Search this list...

	consumer_name	
1	mani	
2	Roy	
3	chinni	
4	john	
5	harry	
6	joe	
7	sanjay	
8	joe	
9	Nani	

Gmail YouTube Maps News Translate All Bookmarks

LAPTOP RENTALS Total Laptops consumer Laptop Bookings Billing Process Laptop Analytics data analytics of laptops

Search...

New Import Assign Label

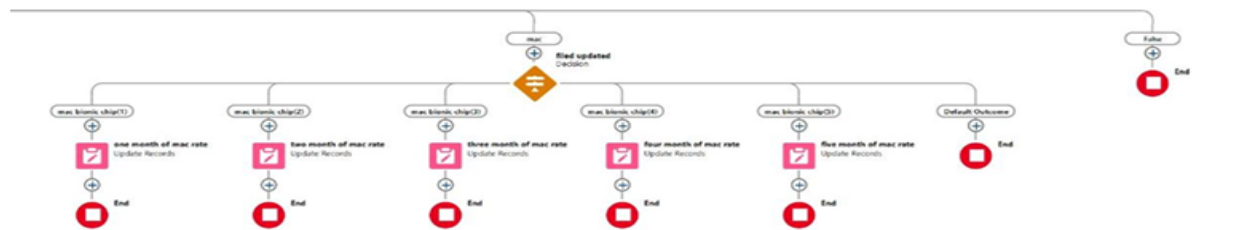
Laptop Bookings Recently Viewed

8 items • Updated a few seconds ago

Search this list...

	Laptop Bookings	
1	Mac Laptop	
2	Hp Laptop	
3	Dell Laptop	
4	Acer Laptop	
5	Acer Laptop	
6	Dell Laptop	
7	Hp Laptop	
8	Mac Laptop	

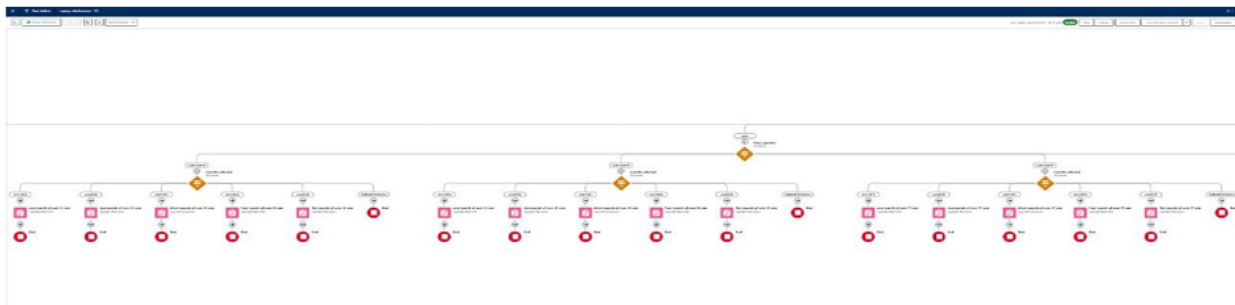
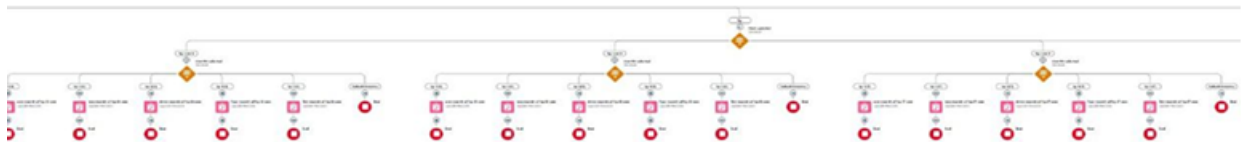
Analytics



Business Logic Design:

- Developed the LaptopBookingHandler class to handle email notifications using **Messaging.SingleEmailMessage**.
- Added the **AfterInsert** trigger on **Laptop_Booking_c** to initiate **sendEmailNotification** method, ensuring customers receive a welcome and confirmation email after booking.

Flow Automation:



Flow Builder Laptop distributions - V3

Last saved on 5/1/2025, 10:50 pm **Active** Run Debug View Tests Save As New Version Save Deactivate

Update Records

*Label: three month of hp i3 rate *API Name: three_month_of_hp_i3_rate

Description:

How to Find Records to Update and Set Their Values

- ☒ Use the laptop bookings record that triggered the flow
- ☐ Update records related to the laptop bookings record that triggered the flow
- ☐ Use the IDs and all field values from a record or record collection
- ☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record: None—Always Update Record

Set Field Values for the Laptop Bookings Record

Field: Amount_c Value: 3000

Validation Rule:

Setup Home Object Manager

SETUP > OBJECT MANAGER

consumer

Validation Rules

1 Items, Sorted by Rule Name

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Phonenumberoremailblankrule	Top of Page	please fill the phone number and email id	✓	PRUDHVI KUMAR BOTTA, 03/01/2025, 5:35 pm

Conditional Formula:

Setup Search Setup

Setup Home Object Manager

consumer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

consumer Validation Rule

Back to consumer

Validation Rule Detail

Rule Name Phonenumeroemailblankrule Edit Clone

Active ☒

Error Condition Formula OR(ISBLANK(Phone_number__c), ISBLANK(Email__c))

Error Message phone number and email should not be blank Error Location Top of Page

Description

Created By Laloramoth Bonela 10/01/2025, 5:46 pm Modified By Laloramoth Bonela 10/01/2025, 5:46 pm

Edit Clone

Apex class:

Created Apex class with name of LaptopBookingHandler

Setup Search Setup

Setup Home Object Manager

apex

Email

Apex Exception Email

Custom Code

Apex Classes

Apex Settings

Apex Test Execution

Apex Test History

Apex Triggers

Environments

Jobs

Apex Flex Queue

Apex Jobs

Didn't find what you're looking for? Try using Global Search.

Apex Classes

Apex Class Detail

Edit Delete Download Security Show Dependencies

Name LaptopBookingHandler

Namespace Prefix

Code Coverage 0% (0/10)

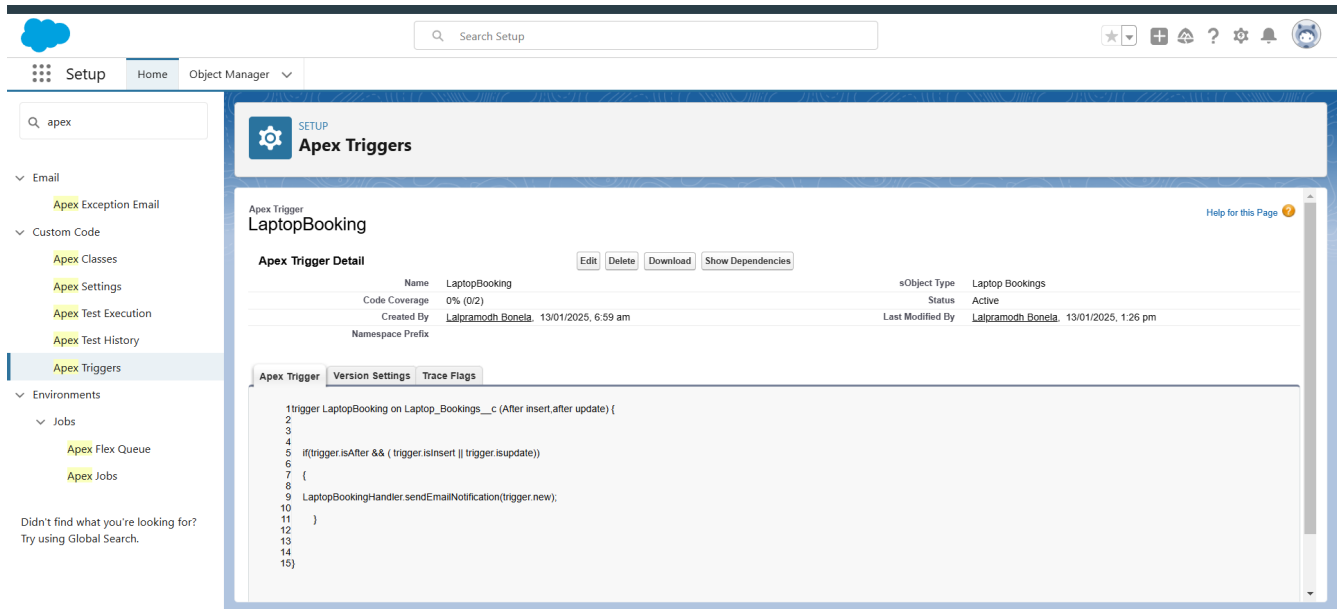
Created By Laloramoth Bonela 13/01/2025, 7:01 am

Last Modified By Laloramoth Bon

Class Body Class Summary Version Settings Trace Flags

```
1 public class LaptopBookingHandler {
2
3   public static void sendEmailNotification (List<Laptop_Bookings__c> lapList){
4     for(Laptop_Bookings__c lap:lapList)
5     {
6
7       Messaging.SingleEmailMessage email = new Messaging.SingleEmailMessage();
8
9       email.setToAddresses( new List<String>({lap.Email__c});
10
11       email.setSubject('Welcome to our company');
12
13       string body = 'Dear Customer, Wt';
14
15       body += 'Welcome to Laptop Rentals! You have been seen as a valuable customer to us. Please continue your journey with us, while we try to provide you with good quality resources. Your Laptop Amount = ' + lap.Amount__c;
16
17       email.setPlainTextBody(body);
18
19       Messaging.sendEmail(new List<Messaging.SingleEmailMessage>({email});
20
21
22 }
```

Apex Triggers:



Created an APEX Trigger with name of **LaptopBooking**

Reports and Dashboards:

Reports and dashboards in Salesforce provide valuable insights into the Laptop rentals CRM application, helping the business monitor bookings, revenue, and customer trends effectively.

Reports:

1. **Tabular Reports:** Used for simple lists, such as a full list of active laptop bookings or consumer records, giving a quick snapshot of data.
2. **Summary Reports:** Offers data with grouped subtotals, such as revenue grouped by laptop type, making it easy to identify high performing categories.

Dashboards:

3. **Revenue Overview:** Shows total rental revenue over time, helping track financial performance.
4. **Popular Laptops:** Displays the most rented laptop types, aiding inventory and marketing decisions.
5. **Customer Insights:** Provides data on customer types, helping target future marketing campaigns more effectively.

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⚙️

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👤

LAPTOP RENTALS

Total Laptops

consumer

Laptop Bookings

Billing Process

* Laptop Analytics

* data analytics of laptops

Report: consumer with Laptop Bookings and Total Laptops

Enable Field Editing

+

Add Chart

▼

🔄

Edit

▼

Total Records

Total Amount

Total Laptops Available

Total Total No Of Laptops: Laptop...

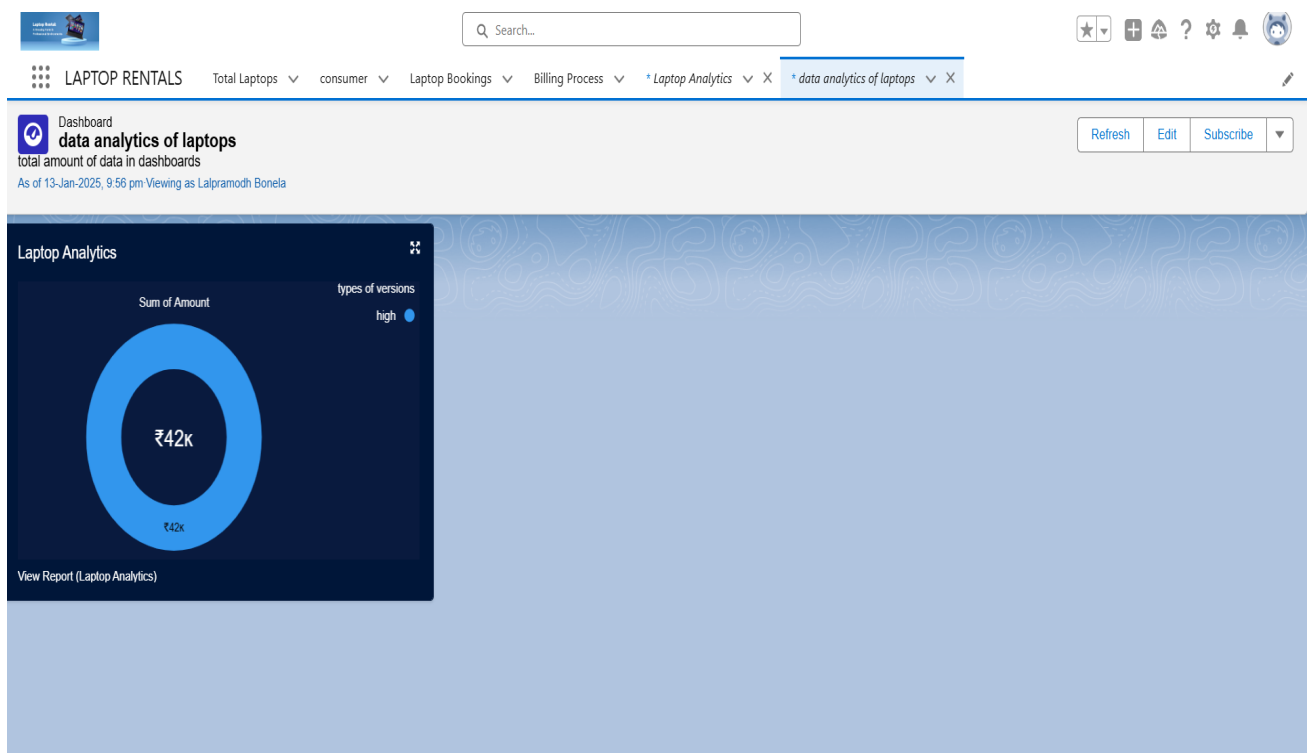
8

₹41,700

390

342

types of versions	consumer: consumer_name	Laptop Bookings: Laptop Bookings	Total No Of Laptops: Total Laptops	Amount	Laptops Available	Total No Of Laptops: Laptops Available
high (8)	Roy	Dell Laptop	2	₹10,000	48	48
	Nani	Acer Laptop	2	₹3,500	49	49
	joe	Mac Laptop	1	₹4,600	49	49
	sanjay	Hp Laptop	2	₹5,000	48	48
	harry	Dell Laptop	4	₹4,200	49	49
	john	Mac Laptop	3	₹6,000	49	49
	chinni	Hp Laptop	5	₹5,900	49	49
	mani	Acer Laptop	1	₹2,500	49	49
Subtotal				₹41,700	390	342
Total (8)				₹41,700	390	342



4. Testing and Validation:

Unit Testing:

Each Apex class and trigger was thoroughly tested, ensuring at least 75% code coverage. Tests were designed to validate that:

- i. Booking confirmation emails are sent correctly.
- ii. Inventory statuses update accurately.

User Interface Testing:

Verified that the UI elements display correctly on page layouts and that all required fields and related lists are accessible. Ensured users could access only the fields allowed by their profile permissions.

Test Scenarios:

Scenario 1: Create a new booking and verify that an email is sent with the correct booking details.

Scenario 2:

Check that validation rules enforce required fields like Email__c and Amount__c.

Scenario 3:

Test that inventory adjust automatically when a booking status is changed from "Booked" to "Returned."

5. Key Scenarios Addressed by Salesforce in the Implementation Project:

This project addresses several critical scenarios to ensure a smooth rental experience for both customers and staff:

● **Automated Notifications:**

Automatically send confirmation emails when a booking is created and provide status updates as the rental progresses. This keeps customers informed and reduces the workload for staff.

● **Real-Time Inventory Management:**

Track the availability of laptops in real time, updating inventory counts when laptops are rented and returned.

● **Workflow Automation for Booking Lifecycle:**

Manage booking statuses with automation, moving each booking from "Booked" to "In

Use" and finally "Returned" without manual intervention.

- **Role-Based_Data_Access:**

Define roles and permissions, allowing staff to manage bookings and customers, while customers can only access their rental information.

6. Conclusion:

The **Laptop Rentals CRM** application demonstrates an effective use of Salesforce to automate and manage key rental business processes. Leveraging custom objects, Apex programming, and flow automation, the project streamlines data management and enhances customer interactions. Key features like email notifications, dynamic pricing, and streamlined booking processes have improved both customer experience and operational efficiency.

Salesforce's reporting and dashboard tools offer valuable insights into rental trends, revenue tracking, and decision-making. With validation rules and role-based access, the solution ensures data security and integrity, supporting sustainable business growth. This project not only meets immediate objectives but also establishes a scalable foundation for future enhancements, positioning Laptop Rentals for long-term success in customer satisfaction and operational efficiency.